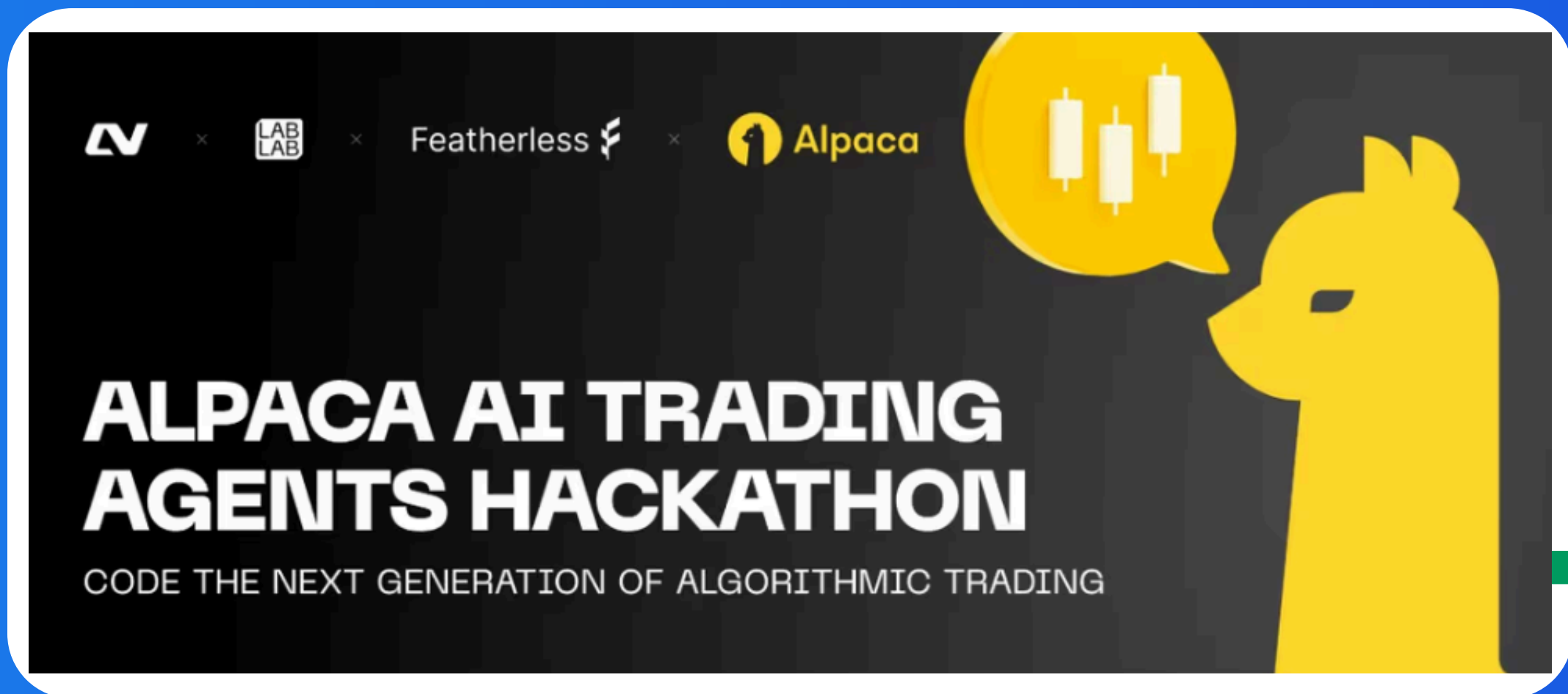


ALPACA AI TRADING HACKATHON



AI hackathon project done by SK AMAN AKMAL



AUTONOMOUS MULTI-AGENT OPTIONS TRADING SYSTEM

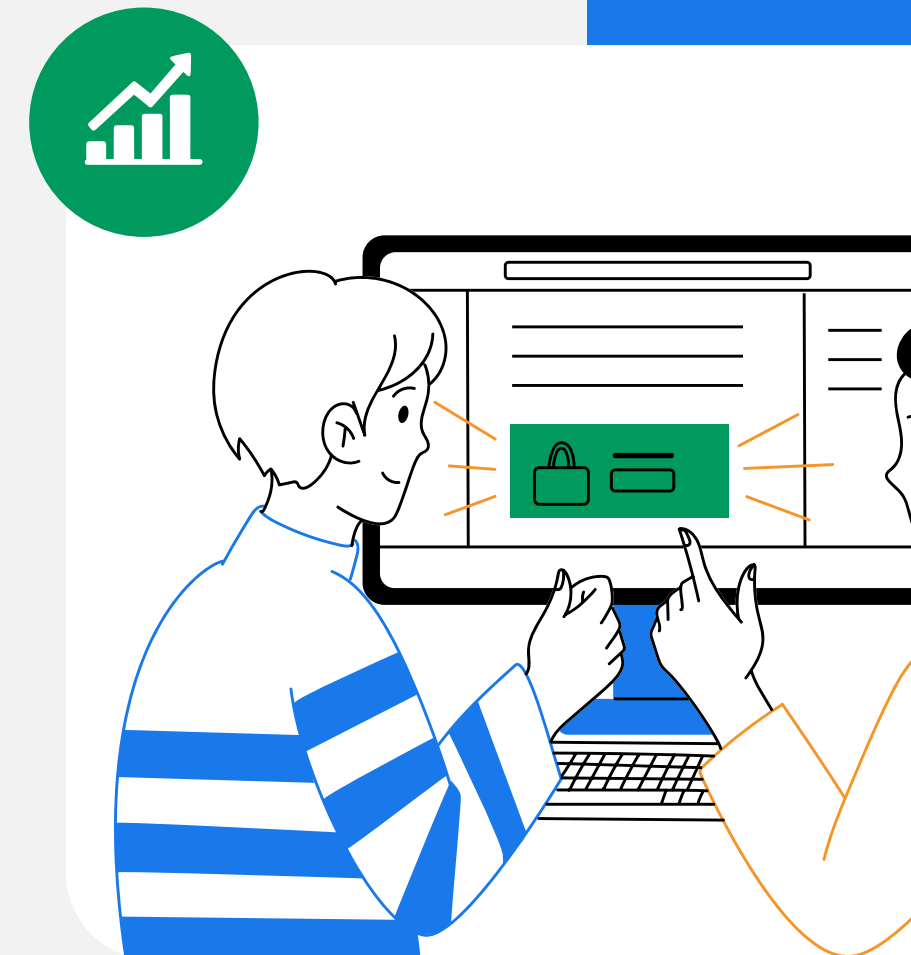
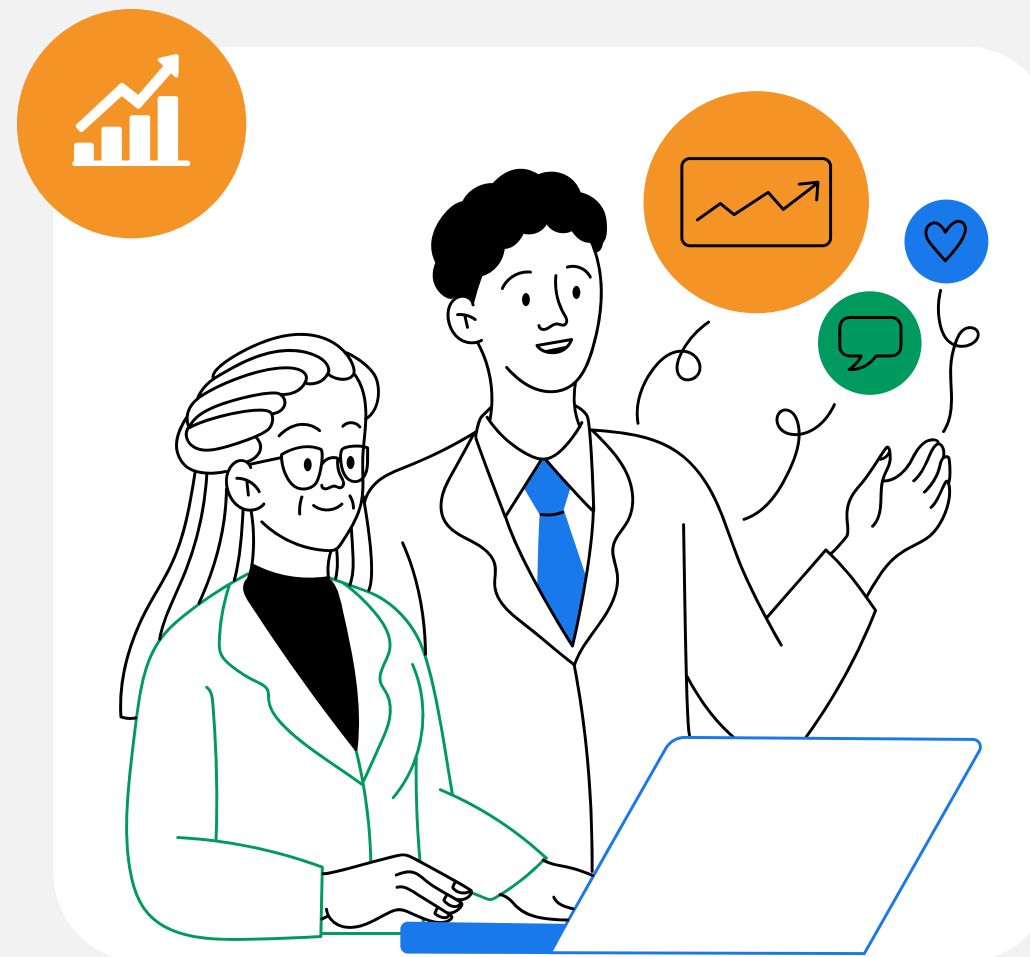
QUANTITATIVE VARIANCE RISK PREMIUM (VRP) ALPHA WITH ZERO-HALLUCINATION COMPLIANCE

Problem:

- Standard LLMs hallucinate risk parameters, misjudge options Greeks, and risk catastrophic portfolio drawdowns with unhedged naked options.
- Institutional finance demands strict, mathematical safety guarantees before any order touches a broker

[READ MORE](#)

Lablab.ai / Alpaca AI Trading Hackathon.



3-LAYER DECOUPLED ARCHITECTURE



LAYER 1 : Primary Options Alpha Agent (alpha_agent.py). -----

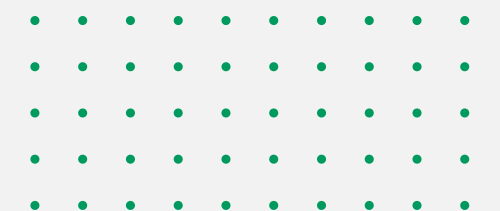
- Variance Risk Premium Calculation: $VRP = \text{Implied Volatility} - \text{Realized Volatility}$
- Featherless AI (Qwen/Qwen2.5-7B-Instruct) for market & volatility regime extraction.

LAYER 2: Deterministic Compliance Shield (compliance_shield.py). -----

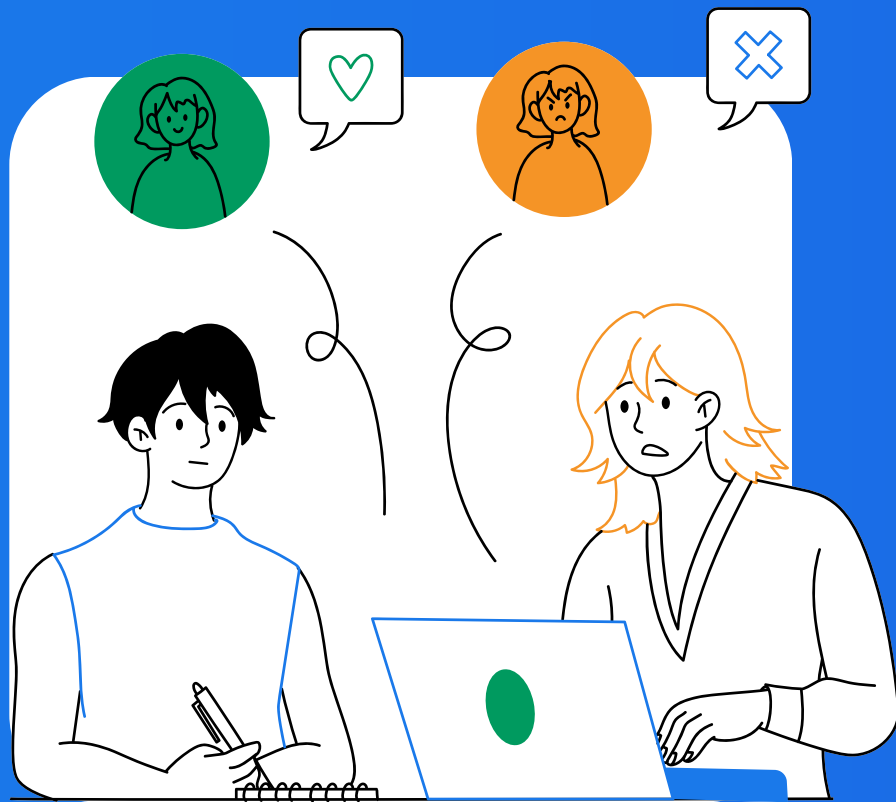
- Zero-LLM, mathematical risk firewall auditing 8 strict portfolio gates.

LAYER3: Alpaca Options Execution Engine (execution_engine.py). -----

- Routes defined-risk multi-leg orders directly to Alpaca Level 3 Options API.



THE 8-GATE COMPLIANCE SHIELD



- 1. STRICT PAPER ENDPOINT: REJECTS ANY LIVE EXECUTION ATTEMPT.
- 2. 5% CAPITAL CEILING: MAX TRADE LOSS CAPPED AT $\leq \$5,000$.
- 3. MARGIN CHECK: REQUIRED COLLATERAL $\leq$ AVAILABLE BUYING POWER.
- 4. DEFINED-RISK WINGS ONLY: ZERO NAKED OPTIONS ALLOWED.
- 5. DTE SAFETY BAND: 7 TO 45 DAYS TO EXPIRATION (NO 0-DTE GAMMA RISK).
- 6. PROBABILITY DELTA BAND: $0.15 \leq |\text{DELTA}| \leq 0.30$ (~80% WIN PROBABILITY).
- 7. LIQUIDITY SPREAD FILTER: BID-ASK SPREAD $\$ \setminus \text{LE } 20 \setminus \% \$$.
- 8. CONCENTRATION LIMIT: CAPPED AT $\$ < 5\$$ ACTIVE POSITIONS.
- **RULE: IF ANY SINGLE GATE FAILS, THE TRADE IS INSTANTLY VETOED.**

ALPACA AI TRADING PLATFORM

FEATHERLESS AI

LAB LAB.AI

Asset	Order Type	Side	Qty	Filled Qty	Avg. Fill Price	Status
<u>2-Leg Order</u>	Limit @ \$0.18	-	1.00	0.00	-	accepted
<u>QQQ260918P00695000</u>	Limit @ -	sell	1.00	0.00	-	accepted
<u>QQQ260918P00693000</u>	Limit @ -	buy	1.00	0.00	-	accepted

order ID - 663e60a9-495c-471b-b296-f5afabbc64a5

test run acc

\$100,000.00

Paper Account PA3ER9YCTD6I

0.00%



```

>>> Analyzing Underlying: SPY <<<
[1/5] Fetching historical bars for SPY...
[1/5] Realized Vol (300): 11.73%, Spot: $773.12
[2/5] Fetching option contracts for SPY around spot $773.12...
[2/5] Retrieved 500 contracts within strike range $718-$827.
[3/5] Fetching indicative quotes for SPY260918C00791000 and SPY260918C00793000...
[3/5] Option quotes retrieved successfully.
[4/5] Short IV: 9.90%, Realized Vol: 11.73%, VRP Spread: -1.83%
[5/5] Querying Featherless AI for market regime analysis...
[5/5] Featherless Regime: VOLATILITY_HARVESTING
Proposed Strategy: BEAR_CALL_SPREAD on SPY
VRP Spread: -1.83% (IV: 9.90%, RV: 11.73%)
Short Leg: SPY260918C00791000 (Strike: $791.0, Delta: 0.1399)
Long Wing: SPY260918C00793000 (Strike: $793.0, Delta: 0.1091)
Max Defined Loss: $173.00 | Expected Credit: $27.00
AI Rationale: The IV is lower than the RV, indicating an opportunity to harvest volatility by selling credit spreads.

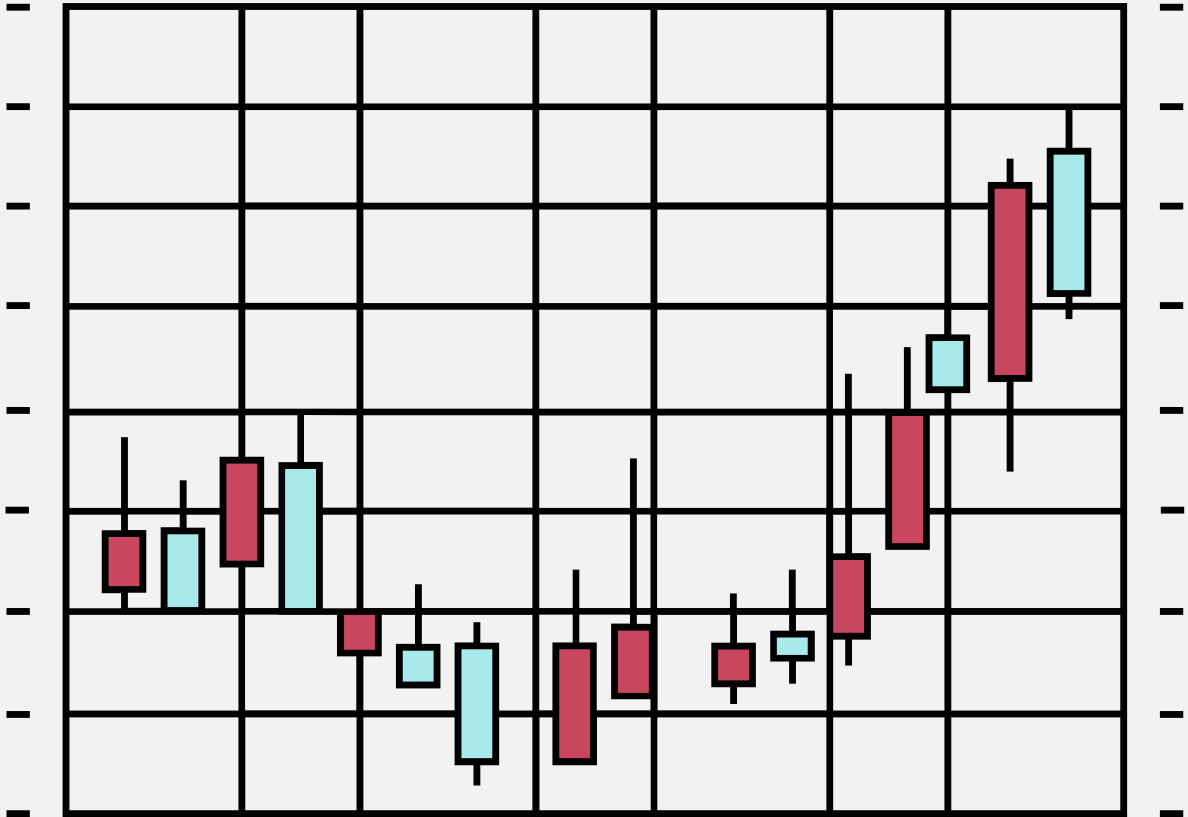
Routing to Compliance Shield...
[COMPLIANCE VETO] Trade on SPY halted by shield: COMPLIANCE SHIELD VETO: Trade VRP-SPY-1788524729 rejected. Violations: Sho
>>> Analyzing Underlying: QQQ <<<
[1/5] Fetching historical bars for QQQ...

```

StandardPremiumDeluxe

Features

ValueSupportUsabilityPerformance



CONCLUSION & ARCHITECTURE HIGHLIGHTS

- **Deterministic Risk Guarantees + Open-Source LLM Alpha**
- **Powered by: Alpaca Options API & Featherless AI**
- **GitHub Repository: Open source, modular, and fully reproducible.**

